

Groups and Lie Algebras: Solutions to Exercise Sheet 6

1 Examples of Lie groups

Problem 1. Show that $\mathrm{GL}(n, \mathbb{R})$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, and $\mathrm{SU}(n)$ are Lie groups. You may use the implicit function theorem: if $F : M \rightarrow N$ is smooth and $n \in N$ is a regular value, then $F^{-1}(n)$ is a submanifold of M .

Solution. We treat the examples one by one. In each case we need two ingredients:

- (i) the set is a smooth manifold;
- (ii) the group multiplication and inverse maps are smooth.

For matrix groups, the second point is usually straightforward because matrix multiplication and matrix inversion are smooth functions of the matrix entries where they are defined.

1. The general linear group $\mathrm{GL}(n, \mathbb{R})$

Recall that

$$\mathrm{GL}(n, \mathbb{R}) = \{A \in \mathrm{Mat}_n(\mathbb{R}) \mid \det A \neq 0\}.$$

The ambient space $\mathrm{Mat}_n(\mathbb{R})$ is naturally isomorphic to $\mathbb{R}^{n^2}$ ($A \mapsto (A_{11}, A_{12}, \dots, A_{nn})$), so it is a smooth manifold.

The determinant map

$$\det : \mathrm{Mat}_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

is a polynomial in the matrix entries, hence is smooth. Since

$$\mathrm{GL}(n, \mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\}),$$

and $\mathbb{R} \setminus \{0\}$ is open in $\mathbb{R}$, it follows that $\mathrm{GL}(n, \mathbb{R})$ is an open subset of $\mathrm{Mat}_n(\mathbb{R})$. Therefore $\mathrm{GL}(n, \mathbb{R})$ is a smooth manifold of real dimension

$$\dim \mathrm{GL}(n, \mathbb{R}) = n^2.$$

The group multiplication

$$\mathrm{GL}(n, \mathbb{R}) \times \mathrm{GL}(n, \mathbb{R}) \longrightarrow \mathrm{GL}(n, \mathbb{R}), \quad (A, B) \mapsto AB$$

is smooth because the entries of AB are polynomial functions of the entries of A and B . The inverse map

$$\mathrm{GL}(n, \mathbb{R}) \longrightarrow \mathrm{GL}(n, \mathbb{R}), \quad A \mapsto A^{-1}$$

is smooth because

$$A^{-1} = \frac{1}{\det A} \mathrm{adj}(A),$$

and the entries of $\mathrm{adj}(A)$ are polynomial functions of the entries of A , while $1/\det A$ is smooth on $\mathrm{GL}(n, \mathbb{R})$. Hence $\mathrm{GL}(n, \mathbb{R})$ is a Lie group.

2. The orthogonal group $O(n)$

The orthogonal group is

$$O(n) = \{A \in GL(n, \mathbb{R}) \mid A^T A = I\}.$$

Equivalently, it consists of matrices preserving the standard Euclidean inner product:

$$\langle Ax, Ay \rangle = \langle x, y \rangle.$$

To show that $O(n)$ is a smooth manifold, define

$$F : \text{Mat}_n(\mathbb{R}) \longrightarrow \text{Sym}_n(\mathbb{R}), \quad F(A) = A^T A,$$

where $\text{Sym}_n(\mathbb{R})$ is the vector space of symmetric $n \times n$ matrices. Then

$$O(n) = F^{-1}(I).$$

We show that I is a regular value of F . Let $A \in O(n)$, and let $X \in T_A \text{Mat}_n(\mathbb{R}) \cong \text{Mat}_n(\mathbb{R})$. Differentiating $(A + tX)^T(A + tX)$ at $t = 0$, we get

$$(DF)_A(X) = X^T A + A^T X.$$

This is a symmetric matrix, as it should be. We need to show that

$$(DF)_A : \text{Mat}_n(\mathbb{R}) \rightarrow \text{Sym}_n(\mathbb{R})$$

is surjective. Let $S \in \text{Sym}_n(\mathbb{R})$. Choose

$$X = \frac{1}{2}AS.$$

Then, since $A^T A = I$ and $S^T = S$,

$$X^T A + A^T X = \frac{1}{2}S^T A^T A + \frac{1}{2}A^T AS = \frac{1}{2}S + \frac{1}{2}S = S.$$

Therefore $(DF)_A$ is surjective for every $A \in F^{-1}(I)$. By the implicit function theorem, $O(n)$ is a smooth submanifold of $\text{Mat}_n(\mathbb{R})$.

The dimension follows from the number of independent equations. The ambient space has dimension n^2 , while $A^T A = I$ imposes a symmetric matrix equation, which gives $n(n+1)/2$ independent conditions. Therefore

$$\dim O(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

The group multiplication and inverse are restrictions of the smooth maps on $GL(n, \mathbb{R})$. Hence $O(n)$ is a Lie group.

3. The special orthogonal group $SO(n)$

The special orthogonal group is

$$SO(n) = \{A \in O(n) \mid \det A = 1\}.$$

Since $A \in O(n)$ satisfies

$$1 = \det(A^T A) = \det(A)^2,$$

we have $\det A = \pm 1$ on $O(n)$. Thus $O(n)$ is the disjoint union of two pieces:

$$O(n) = \{\det = 1\} \sqcup \{\det = -1\}.$$

To see that $SO(n)$ is an embedded submanifold, consider the determinant restricted to $O(n)$:

$$\det : O(n) \longrightarrow \{1, -1\}.$$

Since the image consists only of the two isolated points 1 and -1 , the subset $\{1\} \subset \{1, -1\}$ is both open and closed. Therefore its preimage

$$\det^{-1}(\{1\}) = SO(n)$$

is both open and closed in $O(n)$.

In particular, $SO(n)$ is an open submanifold of $O(n)$. Since open submanifolds have the same dimension as the ambient manifold, we get

$$\dim SO(n) = \dim O(n).$$

From the previous computation for $O(n)$,

$$\dim SO(n) = \dim O(n) = \frac{n(n-1)}{2}.$$

It remains to check that $SO(n)$ is a subgroup. Let $A, B \in SO(n)$. Since $O(n)$ is a group, $AB \in O(n)$. Moreover,

$$\det(AB) = \det(A) \det(B) = 1 \cdot 1 = 1.$$

Hence $AB \in SO(n)$. Also, if $A \in SO(n)$, then $A^{-1} \in O(n)$ and

$$\det(A^{-1}) = (\det A)^{-1} = 1^{-1} = 1.$$

So $A^{-1} \in SO(n)$. Thus $SO(n)$ is a subgroup of $O(n)$.

The group operations on $SO(n)$ are the restrictions of the smooth group operations on $O(n)$. Therefore $SO(n)$ is a Lie group.

4. The unitary group $U(n)$

Now work over $\mathbb{C}$. We regard $\text{Mat}_n(\mathbb{C})$ as a real vector space of dimension $2n^2$. The unitary group is

$$U(n) = \{A \in \text{GL}(n, \mathbb{C}) \mid A^\dagger A = I\},$$

where $A^\dagger = \overline{A}^T$. Equivalently, $U(n)$ consists of complex linear transformations preserving the standard Hermitian inner product.

Define

$$F : \text{Mat}_n(\mathbb{C}) \longrightarrow \text{Herm}_n, \quad F(A) = A^\dagger A,$$

where Herm_n is the real vector space of Hermitian matrices. Then

$$U(n) = F^{-1}(I).$$

At $A \in U(n)$, the differential is

$$(DF)_A(X) = X^\dagger A + A^\dagger X.$$

We show that this is surjective onto Herm_n . Let $H \in \text{Herm}_n$, and choose

$$X = \frac{1}{2}AH.$$

Then

$$X^\dagger A + A^\dagger X = \frac{1}{2}H^\dagger A^\dagger A + \frac{1}{2}A^\dagger AH = \frac{1}{2}H + \frac{1}{2}H = H.$$

Thus I is a regular value, so $U(n)$ is a smooth submanifold of $\text{Mat}_n(\mathbb{C})$.

The real dimension of Herm_n is n^2 : there are n real diagonal entries and $n(n-1)/2$ complex off-diagonal entries. Hence

$$\dim_{\mathbb{R}} U(n) = 2n^2 - n^2 = n^2.$$

The multiplication and inverse maps are smooth as before, so $U(n)$ is a Lie group.

5. The special unitary group $SU(n)$

The special unitary group is

$$SU(n) = \{A \in U(n) \mid \det A = 1\}.$$

For $A \in U(n)$, we have

$$1 = \det(A^\dagger A) = \overline{\det A} \det A = |\det A|^2,$$

so $\det A \in U(1)$. Therefore the determinant restricts to a smooth group homomorphism

$$\det : U(n) \longrightarrow U(1).$$

Then

$$SU(n) = \det^{-1}(1).$$

One can show that $1 \in U(1)$ is a regular value of this determinant map. A concrete way to see this is to compute the derivative at the identity. If $X \in T_I U(n)$, then

$$\left. \frac{d}{dt} \right|_{t=0} \det(I + tX) = \text{tr } X.$$

For $X \in \text{Lie}(U(n)) = T_I U(n)$, the trace $\text{tr } X$ is purely imaginary. The tangent space of $U(1)$ at 1 is $i\mathbb{R}$, and the map $X \mapsto \text{tr } X$ is surjective onto $i\mathbb{R}$, for example by taking

$$X = i\alpha E_{11}, \quad \alpha \in \mathbb{R}.$$

By translating this argument to any point of $U(n)$, the determinant map is a submersion. Hence $SU(n)$ is a smooth submanifold of $U(n)$.

Since $U(n)$ has real dimension n^2 , and $\det A = 1$ imposes one real condition inside $U(n)$, we get

$$\dim_{\mathbb{R}} SU(n) = n^2 - 1.$$

Again multiplication and inversion are smooth, so $SU(n)$ is a Lie group.

2 Tangent spaces at the identity

Problem 2. The Lie groups in the previous problem are all submanifolds of vector spaces of matrices. The tangent space to $F^{-1}(n)$ at m can be identified with the kernel of

$$(DF)_m : T_m M \rightarrow T_n N.$$

Compute the tangent space at the identity in some of the above examples.

Solution. For a matrix Lie group G , its Lie algebra is the tangent space at the identity:

$$\text{Lie}(G) = T_I G.$$

Because our groups are submanifolds of matrix vector spaces, we can compute this by differentiating the defining equations at I .

1. The Lie algebra of $\text{GL}(n, \mathbb{R})$

Since $\text{GL}(n, \mathbb{R})$ is open in $\text{Mat}_n(\mathbb{R})$, its tangent space at every point is just the full matrix space. Therefore

$$T_I \text{GL}(n, \mathbb{R}) = \text{Mat}_n(\mathbb{R}).$$

Thus

$$\mathfrak{gl}(n, \mathbb{R}) = \text{Mat}_n(\mathbb{R}).$$

The Lie bracket is the commutator

$$[X, Y] = XY - YX.$$

2. The Lie algebra of $\text{O}(n)$

The group $\text{O}(n)$ is defined by

$$A^T A = I.$$

Let $A(t)$ be a smooth curve in $\text{O}(n)$ with

$$A(0) = I, \quad \dot{A}(0) = X.$$

Since $A(t)^T A(t) = I$ for all t , differentiating at $t = 0$ gives

$$\dot{A}(0)^T A(0) + A(0)^T \dot{A}(0) = 0.$$

Using $A(0) = I$, this becomes

$$X^T + X = 0.$$

Thus

$$\mathfrak{o}(n) = T_I \text{O}(n) = \{X \in \text{Mat}_n(\mathbb{R}) \mid X^T + X = 0\}.$$

In words, $\mathfrak{o}(n)$ consists of skew-symmetric real matrices.

Its dimension is the number of independent entries above the diagonal:

$$\dim \mathfrak{o}(n) = \frac{n(n-1)}{2}.$$

For example, for $n = 3$, every element of $\mathfrak{o}(3)$ has the form

$$X = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}, \quad \omega_1, \omega_2, \omega_3 \in \mathbb{R}.$$

This is the infinitesimal version of rotations in three dimensions.

3. The Lie algebra of $\mathrm{SO}(n)$

Since $\mathrm{SO}(n)$ is the identity component of $\mathrm{O}(n)$, it has the same tangent space at the identity. Therefore

$$\mathfrak{so}(n) = T_I \mathrm{SO}(n) = \mathfrak{o}(n).$$

Explicitly,

$$\mathfrak{so}(n) = \{X \in \mathrm{Mat}_n(\mathbb{R}) \mid X^T + X = 0\}.$$

One can also see this by differentiating the determinant condition. If $A(t) \in \mathrm{SO}(n)$, then $\det A(t) = 1$. Differentiating at I , we get

$$\mathrm{tr} X = 0.$$

But every skew-symmetric matrix already has trace zero, so this imposes no new condition beyond $X^T + X = 0$. Hence $\mathfrak{so}(n) = \mathfrak{o}(n)$.

4. The Lie algebra of $\mathrm{U}(n)$

The group $\mathrm{U}(n)$ is defined by

$$A^\dagger A = I.$$

Let $A(t) \in \mathrm{U}(n)$ be a smooth curve with

$$A(0) = I, \quad \dot{A}(0) = X.$$

Differentiating

$$A(t)^\dagger A(t) = I$$

at $t = 0$, we get

$$\dot{A}(0)^\dagger A(0) + A(0)^\dagger \dot{A}(0) = 0.$$

Thus

$$X^\dagger + X = 0.$$

Therefore

$$\mathfrak{u}(n) = T_I \mathrm{U}(n) = \{X \in \mathrm{Mat}_n(\mathbb{C}) \mid X^\dagger + X = 0\}.$$

These are called skew-Hermitian matrices.

Notice that $\mathfrak{u}(n)$ is a real vector space, not a complex vector space. Indeed, if $X^\dagger + X = 0$, then generally $(iX)^\dagger + iX \neq 0$. The real dimension is

$$\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2.$$

5. The Lie algebra of $\mathrm{SU}(n)$

For $\mathrm{SU}(n)$, we add the condition

$$\det A = 1.$$

Let $A(t) \in \mathrm{SU}(n)$ with $A(0) = I$ and $\dot{A}(0) = X$. We already know from unitarity that

$$X^\dagger + X = 0.$$

Now differentiate $\det A(t) = 1$. The derivative of the determinant at the identity is the trace:

$$\left. \frac{d}{dt} \right|_{t=0} \det(I + tX) = \mathrm{tr} X.$$

Hence

$$\operatorname{tr} X = 0.$$

Therefore

$$\mathfrak{su}(n) = T_I \operatorname{SU}(n) = \{X \in \operatorname{Mat}_n(\mathbb{C}) \mid X^\dagger + X = 0, \operatorname{tr} X = 0\}.$$

So $\mathfrak{su}(n)$ consists of traceless skew-Hermitian matrices. Its real dimension is

$$\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1.$$

Summary table

Lie group G	$\operatorname{Lie}(G) = T_I G$	dimension
$\operatorname{GL}(n, \mathbb{R})$	$\operatorname{Mat}_n(\mathbb{R})$	n^2
$\operatorname{O}(n)$	$\{X \mid X^T + X = 0\}$	$\frac{n(n-1)}{2}$
$\operatorname{SO}(n)$	$\{X \mid X^T + X = 0\}$	$\frac{n(n-1)}{2}$
$\operatorname{U}(n)$	$\{X \mid X^\dagger + X = 0\}$	n^2
$\operatorname{SU}(n)$	$\{X \mid X^\dagger + X = 0, \operatorname{tr} X = 0\}$	$n^2 - 1$

3 $\operatorname{SO}(3)$ representations, spherical harmonics, and the hydrogen atom

Problem 3. The spherical harmonics are given by

$$Y_l^m(\theta, \varphi) = e^{im\varphi} P_l^m(\cos \theta) N_{lm},$$

where P_l^m is an associated Legendre polynomial and $m \in \{-l, \dots, l\}$. For fixed l , the spherical harmonics span a subspace $\mathcal{H}^{(l)} \subset L^2(S^2)$.

- (a) For fixed l , determine the matrix $D_{mm'}^{(l)}(R)$ for a rotation by an angle α around the z -axis.
- (b) This shows that $\mathcal{H}^{(l)}$ is a representation of $\operatorname{SO}(3)$. Is it irreducible?
- (c) Compute the action of the Hamiltonian for the hydrogen atom on $L^2(\mathbb{R}) \otimes \mathcal{H}^{(l)}$. We know from the lecture that it commutes with the action of $\operatorname{SO}(3)$, hence it preserves these subspaces.

Solution. We recall first how rotations act on functions on the sphere. If $R \in \operatorname{SO}(3)$, the corresponding operator on functions is usually defined by pullback along the inverse rotation:

$$(\rho(R)f)(x) = f(R^{-1}x), \quad x \in S^2.$$

This convention ensures that

$$\rho(R_1)\rho(R_2) = \rho(R_1 R_2).$$

The subspace $\mathcal{H}^{(l)}$ is

$$\mathcal{H}^{(l)} = \operatorname{span}\{Y_l^m \mid m = -l, \dots, l\}.$$

It has dimension $2l + 1$.

Part (a): rotation around the z -axis

Let $R_z(\alpha)$ be the rotation by angle α around the z -axis. In spherical coordinates it acts by

$$(\theta, \varphi) \mapsto (\theta, \varphi + \alpha).$$

Therefore the inverse rotation acts by

$$R_z(\alpha)^{-1} : (\theta, \varphi) \mapsto (\theta, \varphi - \alpha).$$

With the pullback convention, we get

$$(\rho(R_z(\alpha))Y_l^m)(\theta, \varphi) = Y_l^m(\theta, \varphi - \alpha).$$

Using

$$Y_l^m(\theta, \varphi) = N_{lm} e^{im\varphi} P_l^m(\cos \theta),$$

we find

$$\begin{aligned} Y_l^m(\theta, \varphi - \alpha) &= N_{lm} e^{im(\varphi - \alpha)} P_l^m(\cos \theta) \\ &= e^{-im\alpha} Y_l^m(\theta, \varphi). \end{aligned}$$

Thus, with the pullback convention,

$$\rho(R_z(\alpha))Y_l^m = e^{-im\alpha} Y_l^m.$$

So the matrix of $R_z(\alpha)$ in the basis

$$Y_l^{-l}, Y_l^{-l+1}, \dots, Y_l^l$$

is diagonal:

$$D^{(l)}(R_z(\alpha)) = \text{diag}(e^{il\alpha}, e^{i(l-1)\alpha}, \dots, e^{-il\alpha}).$$

Equivalently, in index notation,

$$D_{mm'}^{(l)}(R_z(\alpha)) = e^{-im\alpha} \delta_{mm'}.$$

Remark. Some texts use the opposite convention, writing the transformation law using $f(Rx)$ rather than $f(R^{-1}x)$. With that convention the phase becomes $e^{+im\alpha}$. The two conventions differ only by replacing α with $-\alpha$, or equivalently by using the inverse representation.

The main point is that rotations around the z -axis do not mix different m -values. They only multiply Y_l^m by a phase.

Part (b): irreducibility of $\mathcal{H}^{(l)}$

For fixed l , the space

$$\mathcal{H}^{(l)} = \text{span}\{Y_l^m \mid -l \leq m \leq l\}$$

is a representation of $\text{SO}(3)$ of dimension $2l + 1$. This representation is irreducible.

Let us explain why. The infinitesimal generators of rotations are the angular momentum operators L_x, L_y, L_z , or equivalently

$$L_z, \quad L_+ = L_x + iL_y, \quad L_- = L_x - iL_y.$$

They act on spherical harmonics by

$$L_z Y_l^m = \hbar m Y_l^m,$$

and

$$\begin{aligned} L_+ Y_l^m &= \hbar \sqrt{l(l+1) - m(m+1)} Y_l^{m+1}, \\ L_- Y_l^m &= \hbar \sqrt{l(l+1) - m(m-1)} Y_l^{m-1}. \end{aligned}$$

These formulas show that, starting from any nonzero Y_l^m , repeated application of the raising and lowering operators connects all the basis vectors

$$Y_l^{-l}, Y_l^{-l+1}, \dots, Y_l^l.$$

Consequently any nonzero invariant subspace must contain all of $\mathcal{H}^{(l)}$. Therefore $\mathcal{H}^{(l)}$ has no nontrivial invariant subspaces. Hence it is irreducible.

In representation-theoretic language, $\mathcal{H}^{(l)}$ is the spin- l irreducible representation of $\text{SO}(3)$, of dimension $2l+1$. For $\text{SO}(3)$ only integer values of l occur as honest representations. Half-integer spins are representations of the double cover $\text{SU}(2)$, not of $\text{SO}(3)$ itself.

Part (c): the hydrogen Hamiltonian on radial and angular parts

The Hamiltonian for the hydrogen atom is

$$H = -\frac{\hbar^2}{2\mu} \Delta - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r},$$

where μ is the reduced mass and $r = |x|$. The potential depends only on r , so the Hamiltonian is rotationally invariant. Therefore it commutes with $\text{SO}(3)$ and preserves each angular momentum sector

$$L^2(\mathbb{R}_+, r^2 dr) \otimes \mathcal{H}^{(l)}.$$

In spherical coordinates, the Laplacian is

$$\Delta = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \Delta_{S^2},$$

where Δ_{S^2} is the Laplacian on the unit sphere. The spherical harmonics are eigenfunctions of the spherical Laplacian:

$$\Delta_{S^2} Y_l^m = -l(l+1) Y_l^m.$$

Now take a wave function of separated form

$$\psi(r, \theta, \varphi) = R(r) Y_l^m(\theta, \varphi).$$

Then

$$\begin{aligned} \Delta(R(r) Y_l^m(\theta, \varphi)) &= \left[\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) \right] Y_l^m + \frac{R(r)}{r^2} \Delta_{S^2} Y_l^m \\ &= \left[\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{l(l+1)}{r^2} R(r) \right] Y_l^m. \end{aligned}$$

Therefore

$$\begin{aligned} H(R(r) Y_l^m(\theta, \varphi)) &= \left[-\frac{\hbar^2}{2\mu} \left(\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{l(l+1)}{r^2} R(r) \right) \right. \\ &\quad \left. - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} R(r) \right] Y_l^m(\theta, \varphi). \end{aligned}$$

Thus, on the subspace

$$L^2(\mathbb{R}_+, r^2 dr) \otimes \mathcal{H}^{(l)},$$

the Hamiltonian acts as a radial differential operator tensored with the identity on $\mathcal{H}^{(l)}$:

$$H|_{L^2(\mathbb{R}_+, r^2 dr) \otimes \mathcal{H}^{(l)}} = H_l^{\text{rad}} \otimes \text{id}_{\mathcal{H}^{(l)}},$$

where

$$H_l^{\text{rad}} = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right) - \frac{l(l+1)}{r^2} \right] - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r}.$$

The term

$$\frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2}$$

is often called the centrifugal term.

Equivalently, if we set

$$u(r) = rR(r),$$

then the radial equation becomes an equation on $L^2(\mathbb{R}_+, dr)$:

$$H_l^u u(r) = \left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2} - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} \right] u(r).$$

This is the usual one-dimensional radial Schrödinger operator for fixed angular momentum l .

The representation-theoretic meaning is important. Since H commutes with $\text{SO}(3)$, Schur's lemma implies that H acts as the identity on each irreducible angular momentum representation $\mathcal{H}^{(l)}$, and only acts nontrivially on the radial multiplicity space. Therefore the energy eigenvalues do not depend on m . For each fixed l , the angular degeneracy is

$$\dim \mathcal{H}^{(l)} = 2l + 1.$$

This is the familiar magnetic quantum number degeneracy of the hydrogen atom.